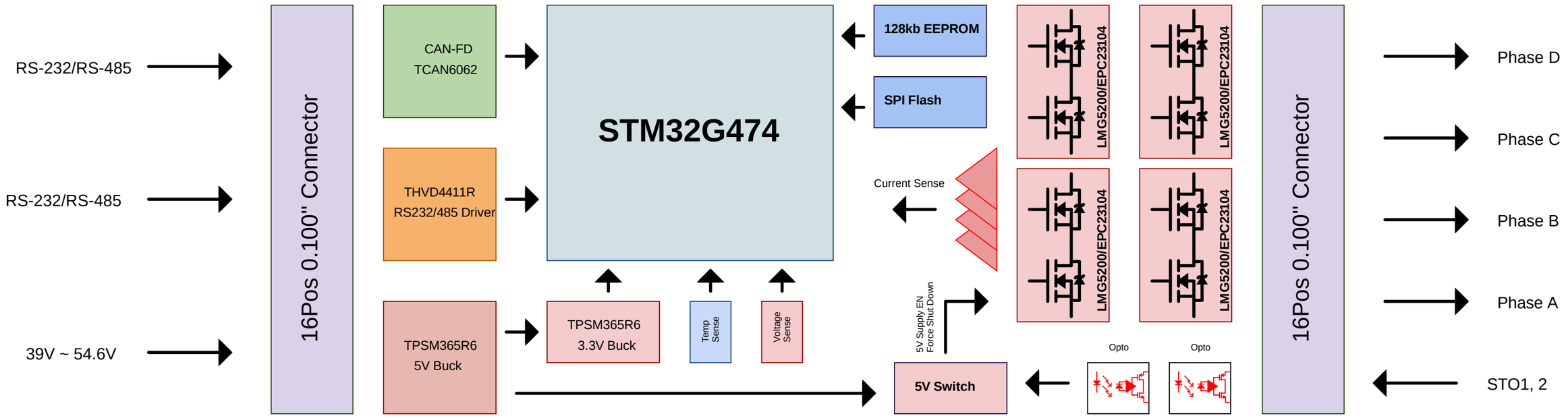
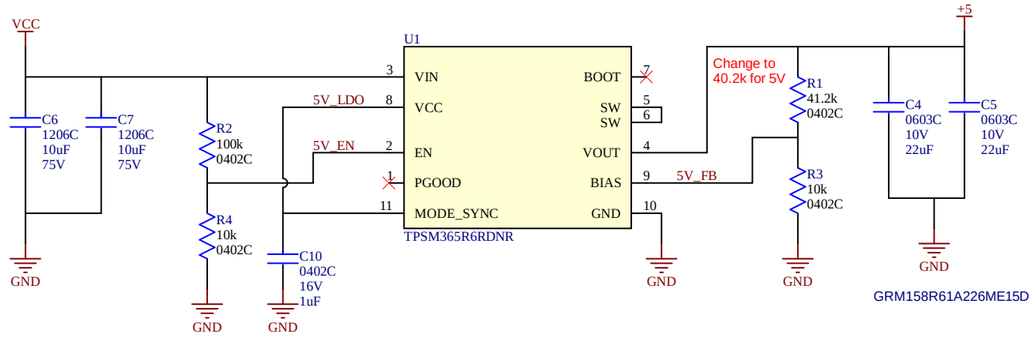
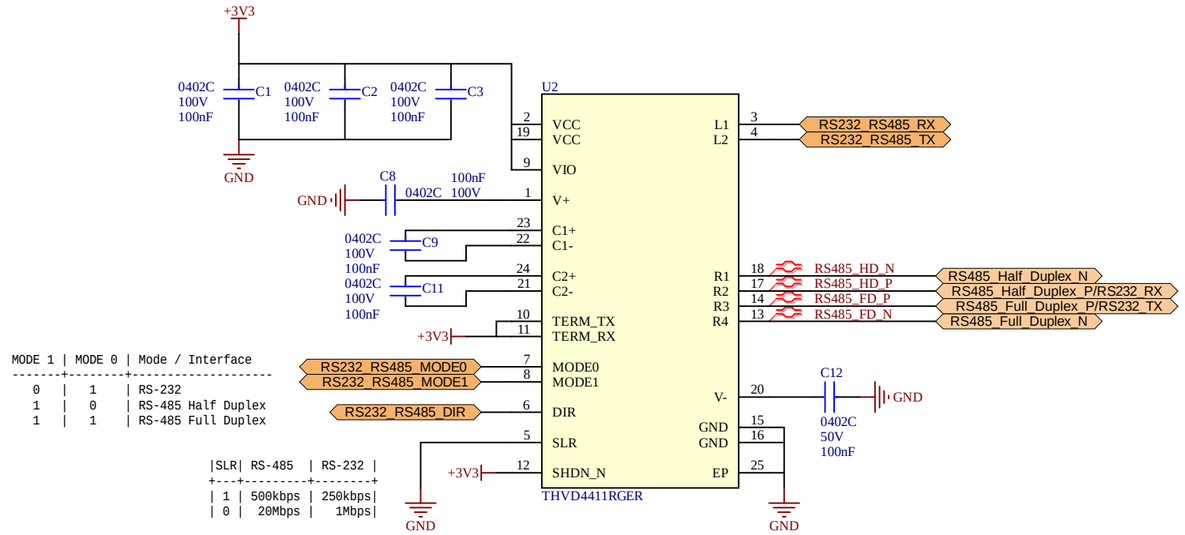


Title: Cover		
Size: A	Number:	Revision:
Date: 11/06/2026		Sheet of
File: D:\Altium\GaN_SoM\test3_cover.SchDoc		Drawn By: Melona Sawara

54.6V ~ 39V → 5V Buck Converter



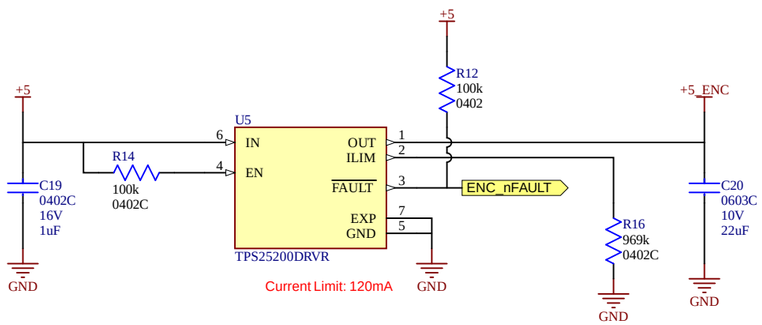
RS-232/RS-485 Driver



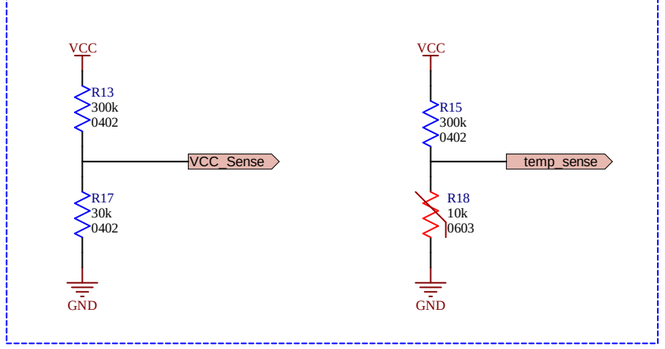
5V → 3.3V Buck Converter

FD-CAN Driver

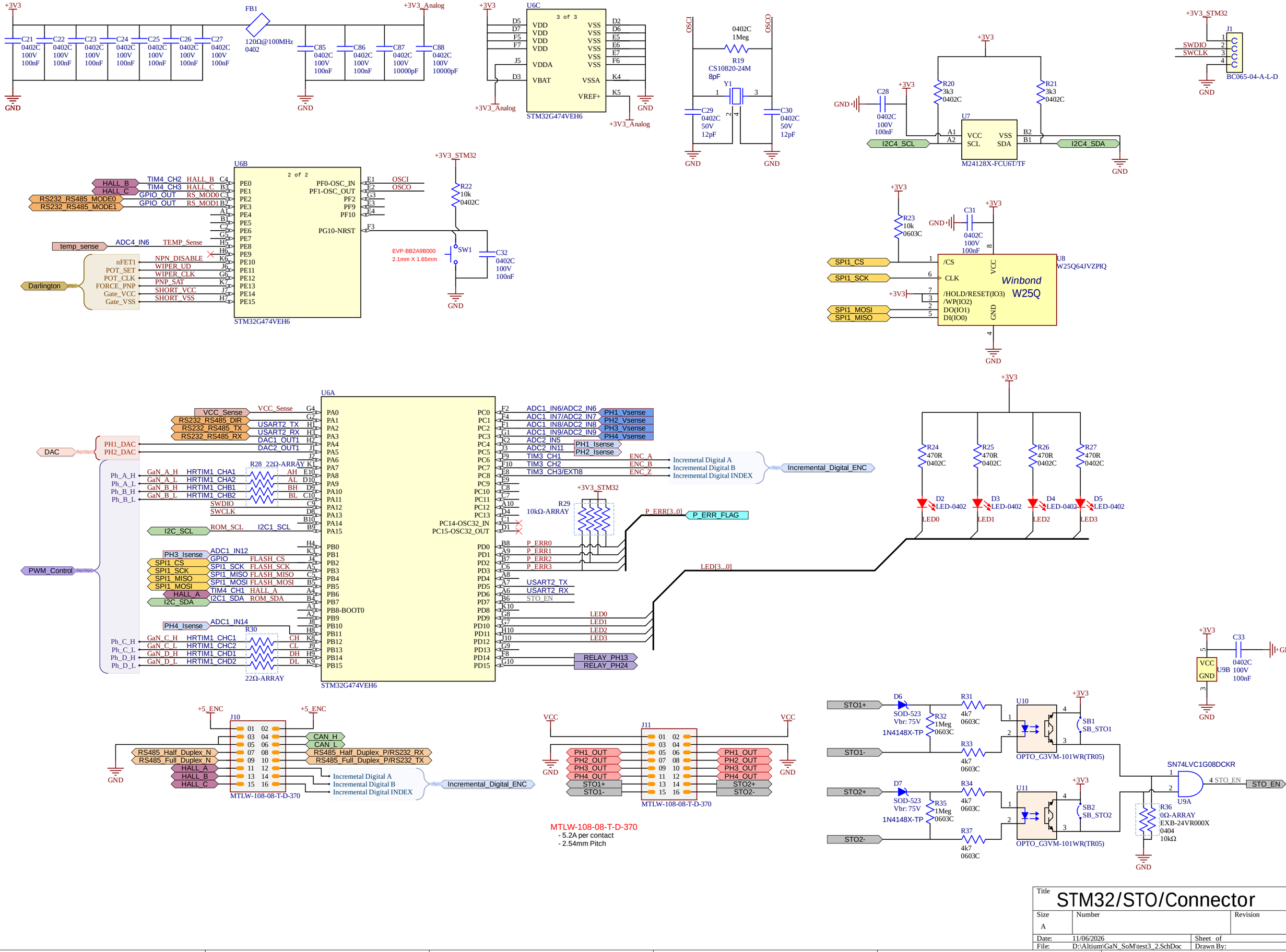
5V Encoder overcurrent Fuse



Temperature/Voltage Sense



Title: Peripheral		
Size: A	Number:	Revision:
Date: 11/06/2026		Sheet of
File: D:\Altium\GaN_SoM\test3_1.1.SchDoc		Drawn By:


+3V3
C21 0402C 100V 100nF
C22 0402C 100V 100nF
C23 0402C 100V 100nF
C24 0402C 100V 100nF
C25 0402C 100V 100nF
C26 0402C 100V 100nF
C27 0402C 100V 100nF
GND
FB1
120Ω@100MHz
0402
+3V3_Analog
C85 0402C 100V 100nF
C86 0402C 100V 100nF
C87 0402C 100V 10000pF
C88 0402C 100V 10000pF
U6C
3 of 3
D5 D7 F5 F7 J5 D3
VDD VDD VDD VDD VDDA VBAT
VSS VSS VSS VSS VSS VSS VSSA VREF+
D2 D6 E5 E6 E7 F6 K4 K5
STM32G474VEH6
OSCI
OSCO
0402C
1Meg
R19
CS10820-24M
8pF
Y1
C29 0402C 50V 12pF
C30 0402C 50V 12pF
+3V3_STM32
J1
SWDIO
SWCLK
BC065-04-A-L-D
R20 3k3 0402C
R21 3k3 0402C
C28 0402C 100V 100nF
U7
VCC SCL VSS SDA
A1 A2 B2 B1
I2C4_SCL
I2C4_SDA
M24128X-FCU6T/TF
U6B
2 of 2
PE0 PE1 PE2 PE3 PE4 PE5 PE6 PE7 PE8 PE9 PE10 PE11 PE12 PE13 PE14 PE15
PF0-OSC_IN PF1-OSC_OUT PF2 PF9 PF10 PG10-NRST
HALL_B HALL_C RS232_RS485_MODE0 RS232_RS485_MODE1
TIM4_CH2 HALL_B
TIM4_CH3 HALL_C
GPIO_OUT RS_MOD0
GPIO_OUT RS_MOD1
temp_sense
ADC4_IN6
TEMP_Sense
Darlington
nFET1 POT_SET POT_CLK FORCE_PNP Gate_VCC Gate_VSS
NPN_DISABLE WIPER_UD WIPER_CLK PNP_SAT SHORT_VCC SHORT_VSS
STM32G474VEH6
R22 10k 0402C
SW1
EVP-BB2A9B000
2.1mm X 1.65mm
C32 0402C 100V 100nF
R23 10k 0603C
C31 0402C 100V 100nF
U8
W25Q64JVZPIQ
SPI1_CS SPI1_SCK SPI1_MOSI SPI1_MISO
/CS CLK /HOLD/RESET(IO3) /WP(IO2) DO(IO1) DI(IO0)
Winbond W25Q
U6A
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8-BOOT0 PB9 PB10 PB11 PB12 PB13 PB14 PB15
PC0 PC1 PC2 PC3 PC4 PC5 PC6 PC7 PC8 PC9 PC10 PC11 PC12 PC13 PC14-OSC32_IN PC15-OSC32_OUT
PD0 PD1 PD2 PD3 PD4 PD5 PD6 PD7 PD8 PD9 PD10 PD11 PD12 PD13 PD14 PD15
VCC_Sense
RS232_RS485_DIR
RS232_RS485_TX
RS232_RS485_RX
USART2_TX
USART2_RX
DAC1_OUT1
DAC2_OUT1
DAC
PH1_DAC
PH2_DAC
R28 22Ω-ARRAY
Ph_A_H Ph_A_L Ph_B_H Ph_B_L
GaN_A_H GaN_A_L GaN_B_H GaN_B_L
HRTIM1_CHA1 HRTIM1_CHA2 HRTIM1_CHB1 HRTIM1_CHB2
SWDIO SWCLK
I2C_SCL
ROM_SCL
I2C1_SCL
PH3_Isense
SPI1_CS SPI1_SCK SPI1_MISO SPI1_MOSI
GPIO FLASH_CS
SPI1_SCK FLASH_SCK
SPI1_MISO FLASH_MISO
SPI1_MOSI FLASH_MOSI
HALL_A
TIM4_CH1 HALL_A
I2C_SDA
I2C1_SDA ROM_SDA
PH4_Isense
ADC1_IN14
R30
Ph_C_H Ph_C_L Ph_D_H Ph_D_L
GaN_C_H GaN_C_L GaN_D_H GaN_D_L
HRTIM1_CHC1 HRTIM1_CHC2 HRTIM1_CHD1 HRTIM1_CHD2
22Ω-ARRAY
PWM_Control
ADC1_IN6/ADC2_IN6
ADC1_IN7/ADC2_IN7
ADC1_IN8/ADC2_IN8
ADC1_IN9/ADC2_IN9
ADC2_IN5
ADC2_IN11
TIM3_CH1
TIM3_CH2
TIM3_CH3/EXTI8
PH1_Vsense
PH2_Vsense
PH3_Vsense
PH4_Vsense
PH1_Isense
PH2_Isense
ENC_A
ENC_B
ENC_Z
Incremental Digital A
Incremental Digital B
Incremental Digital INDEX
Incremental_Digital_ENC
R29 10kΩ-ARRAY
P_ERR[3..0]
P_ERR_FLAG
P_ERR0 P_ERR1 P_ERR2 P_ERR3
USART2_TX
USART2_RX
STO_EN
LED[3...0]
LED0 LED1 LED2 LED3
RELAY_PH13
RELAY_PH24
R24 470R 0402C
R25 470R 0402C
R26 470R 0402C
R27 470R 0402C
D2 LED-0402 LED0
D3 LED-0402 LED1
D4 LED-0402 LED2
D5 LED-0402 LED3
+5_ENC
J10
CAN_H CAN_L
RS485_Half_Duplex_N
RS485_Full_Duplex_N
RS485_Half_Duplex_P/RS232_RX
RS485_Full_Duplex_P/RS232_TX
HALL_A HALL_B HALL_C
MTLW-108-08-T-D-370
J11
PH1_OUT PH2_OUT PH3_OUT PH4_OUT STO1+ STO1-
STO2+ STO2-
MTLW-108-08-T-D-370
- 5.2A per contact
- 2.54mm Pitch
D6 SOD-523 Vbr: 75V 1N4148X-TP
D7 SOD-523 Vbr: 75V 1N4148X-TP
R31 4k7 0603C
R32 1Meg 0603C
R33 4k7 0603C
R34 4k7 0603C
R35 1Meg 0603C
R37 4k7 0603C
U10
U11
OPTO_G3VM-101WR(TR05)
SB1 SB_STO1
SB2 SB_STO2
SN74LVC1G08DCKR
U9A
R36 0Ω-ARRAY EXB-24VR000X 0404 10kΩ
C33 0402C 100V 100nF
U9B
Title
STM32/STO/Connector
Size A
Number
Revision
Date: 11/06/2026
Sheet of
File: D:\Altium\GaN_SoM\test3_2.SchDoc
Drawn By:

Rise Time = (0.3418VIN + 0.1036) × CT + 14.064VIN + 12.255
STO Force Shut Down

+5
C39 0402C 100V 100nF
GND
U12
VIN
VBIAS
ON
VOUT
CT
GND
TPS22995RZFR
STO_EN
tR = 100.7uS
C40 0402C 50V 10pF
C41 0402C 100V 100nF
+5_DRV

SMD Probe Pad
TP1 TP2 TP3 TP4
TP5 TP6 TP7 TP8
PWM_Control
Ph_A_H
Ph_A_L
Ph_B_H
Ph_B_L
Ph_C_H
Ph_C_L
Ph_D_H
Ph_D_L
GaN_A_H
GaN_A_L
GaN_B_H
GaN_B_L
GaN_C_H
GaN_C_L
GaN_D_H
GaN_D_L

1277AS-H-1R5M=P2
AOTA-B201208S1R0MT
L1
Inductor_0805
50mΩ
1uH
PH3_OUT
PH1_OUT
U17
G3VM-101WR(TR05)
RELAY_PH13
R54
470R
0402C
GND

1277AS-H-1R5M=P2
L2
Inductor_0805
50mΩ
1uH
PH4_OUT
PH2_OUT
U20
G3VM-101WR(TR05)
RELAY_PH24
R58
470R
0402C
GND

+5_DRV
R38 0R 0402C
C34 0402C 100V 100nF
CP_HB1
CP_HS1
VCC
HB
HS
VIN
HI
SW
LI
PGND
AGND
U13
LMG5200MOFR
Replace with EPC23104
C43 0402C 50V 10pF
C44 0402C 50V 10pF
R39
5mΩ 0612
R41 EXB-24VR000X 0404 0Ω
PH1_Sense_P
PH1_Sense_N
C78 0402C 50V 1000pF
+3V3
C42 0402C 100V 100nF
U14
INA241A3
20V/V
IN+
IN-
OUT
PH1_Isense
C51 0402C 50V 10nF
R40 24k 0402C
R42 1k 0402C
C49 0402C 50V 10nF
PH1_Vsense
PH1_OUT

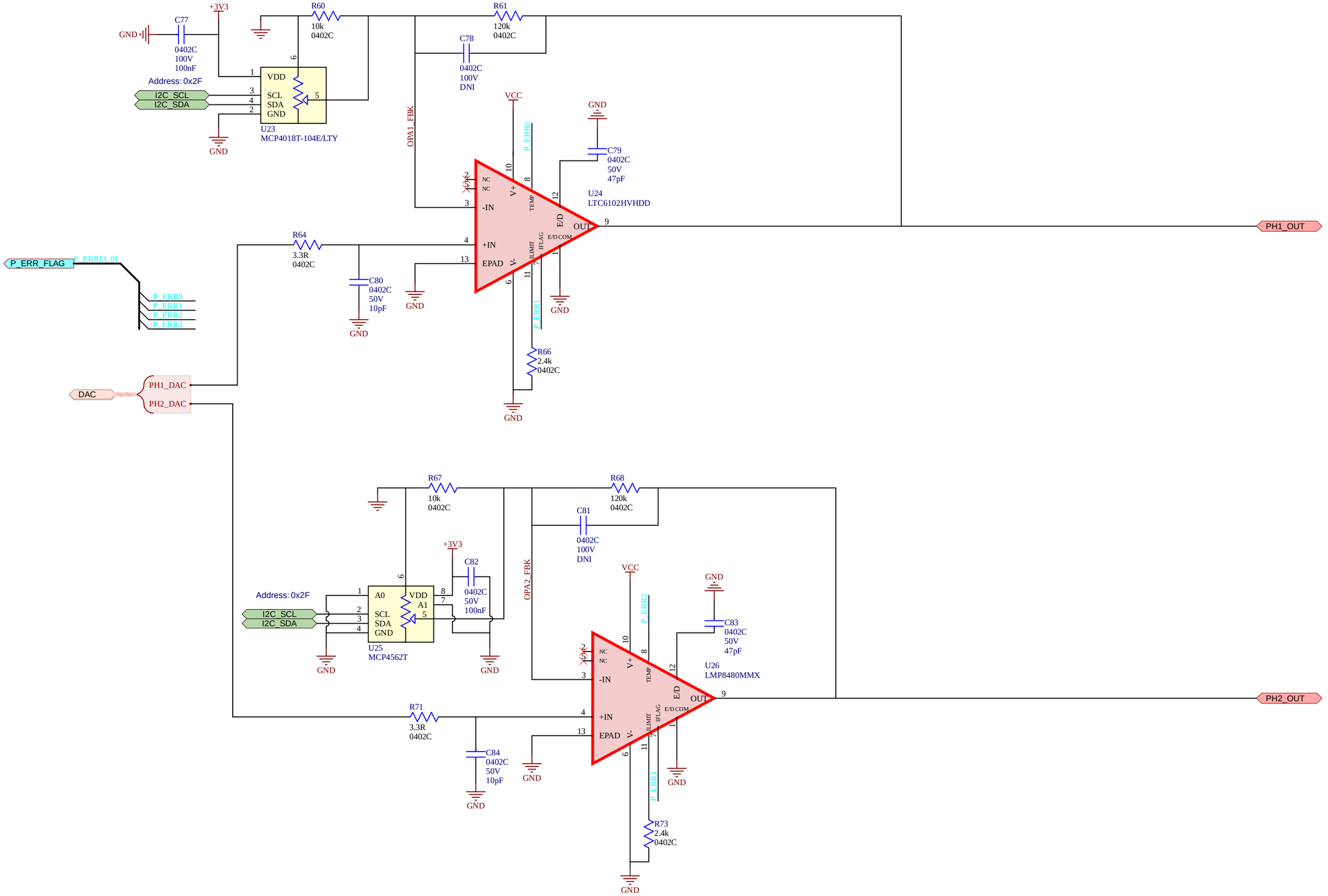
+5_DRV
R43 0R 0402C
C50 0402C 100V 100nF
CP_HB2
CP_HS2
U15
LMG5200MOFR
C57 0402C 50V 10pF
C58 0402C 50V 10pF
R44
5mΩ 0612
R46 EXB-24VR000X 0404 0Ω
PH2_Sense_P
PH2_Sense_N
C89 0402C 50V 1000pF
C56 0402C 100V 100nF
U16
INA241A3
20V/V
C61 0402C 50V 10nF
R45 24k 0402C
R47 1k 0402C
C59 0402C 50V 10nF
PH2_Isense
PH2_Vsense
PH2_OUT

+5_DRV
R48 0R 0402C
C60 0402C 100V 100nF
CP_HB3
CP_HS3
U18
LMG5200MOFR
C67 0402C 50V 10pF
C68 0402C 50V 10pF
R49
5mΩ 0612
R51 EXB-24VR000X 0404 0Ω
PH3_Sense_P
PH3_Sense_N
C90 0402C 50V 1000pF
C66 0402C 100V 100nF
U19
INA241A3
20V/V
C71 0402C 50V 10nF
R50 24k 0402C
R52 1k 0402C
C69 0402C 50V 10nF
PH3_Isense
PH3_Vsense
PH3_OUT

+5_DRV
R53 0R 0402C
C70 0402C 100V 100nF
CP_HB4
CP_HS4
U21
LMG5200MOFR
C73 0402C 50V 10pF
C74 0402C 50V 10pF
R55
5mΩ 0612
R57 EXB-24VR000X 0404 0Ω
PH4_Sense_P
PH4_Sense_N
C91 0402C 50V 1000pF
C72 0402C 100V 100nF
U22
INA241A3
20V/V
C76 0402C 50V 10nF
R56 24k 0402C
R59 1k 0402C
C75 0402C 50V 10nF
PH4_Isense
PH4_Vsense
PH4_OUT

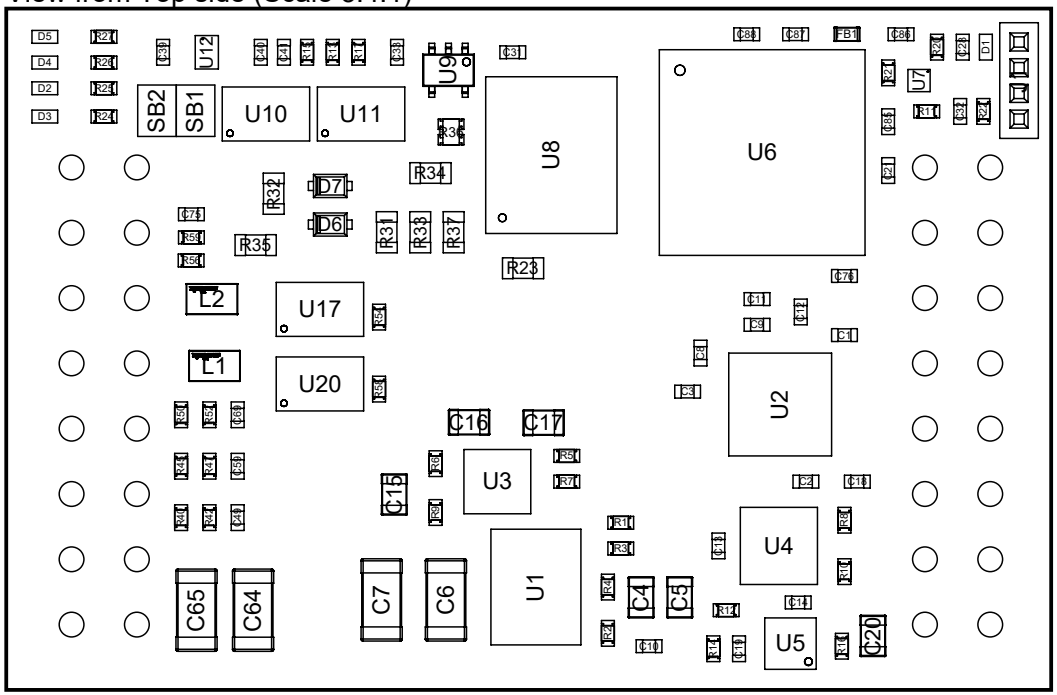
+5_DRV
C35 0402C 100V 100nF
C36 0402C 100V 100nF
C37 0402C 100V 100nF
C38 0402C 100V 100nF
VCC
C45 0402C 100V 330nF
C46 0402C 100V 330nF
C47 0402C 100V 330nF
C48 0402C 100V 330nF
C52 1206C 4u7 100V
C53 1206C 4u7 100V
C54 1206C 4u7 100V
C55 1206C 4u7 100V
C62 1206C 4u7 100V
C63 1206C 4u7 100V
C64 1206C 4u7 100V
C65 1206C 4u7 100V

Title	Half Bridge	
Size: A	Number	Revision
Date: 11/06/2026		Sheet of
File: D:\Altium\GaN_SoM\test3_3.SchDoc		Drawn By:

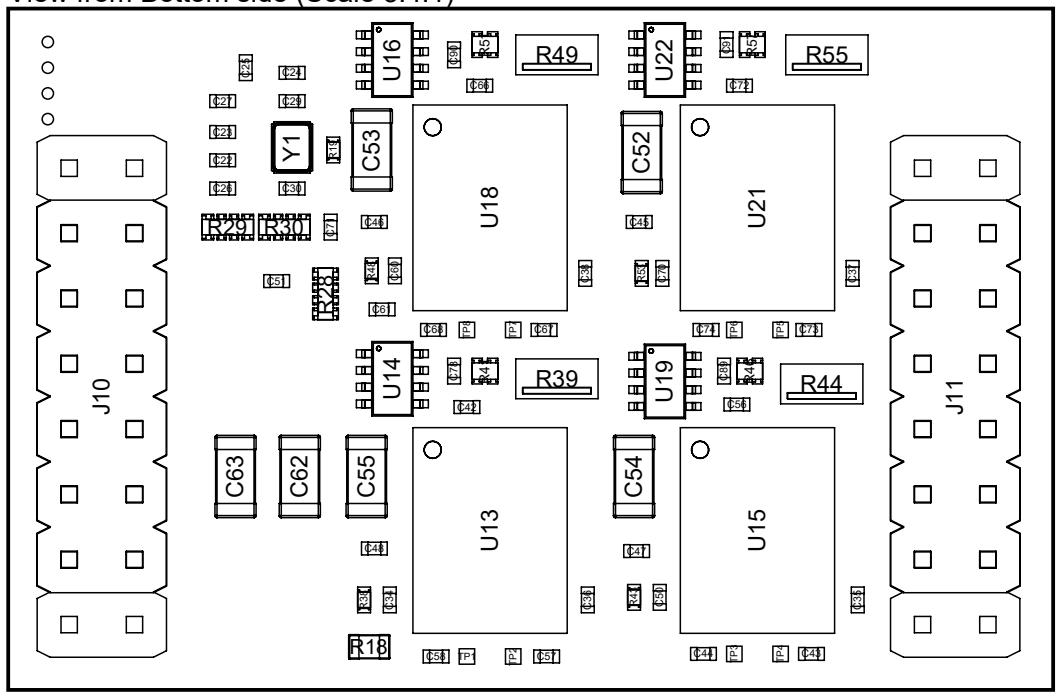


Title	OPA	
Size: A	Number	Revision
Date: 11/06/2026		Sheet of
File: D:\Altium\GaN_SoM\test3_4.SchDoc		Drawn By:

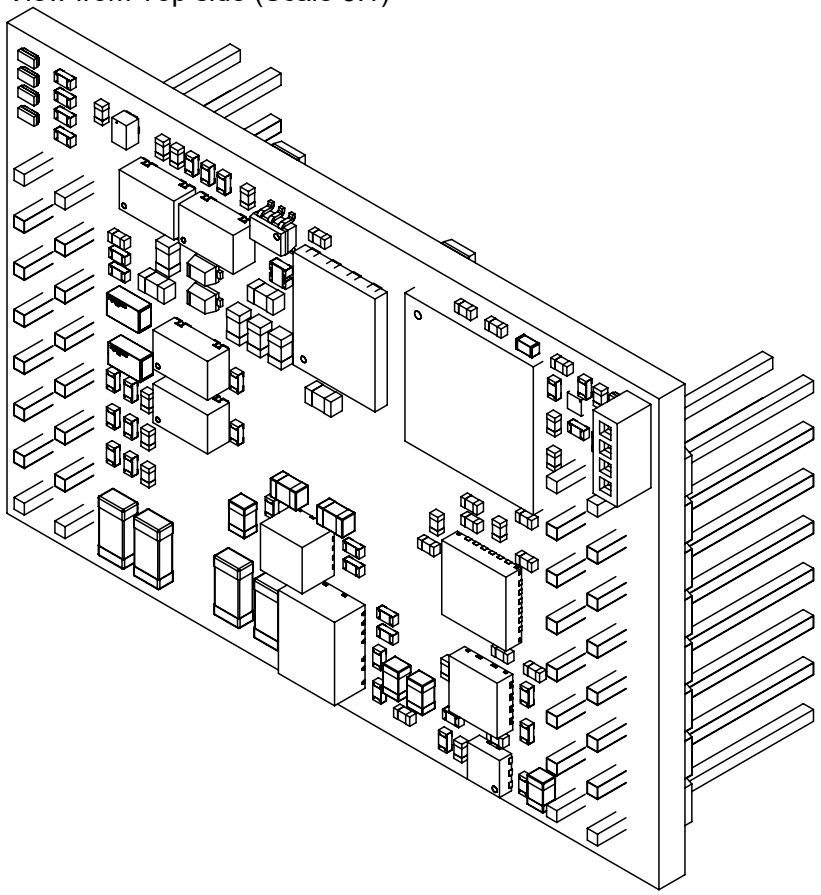
View from Top side (Scale 3.4:1)



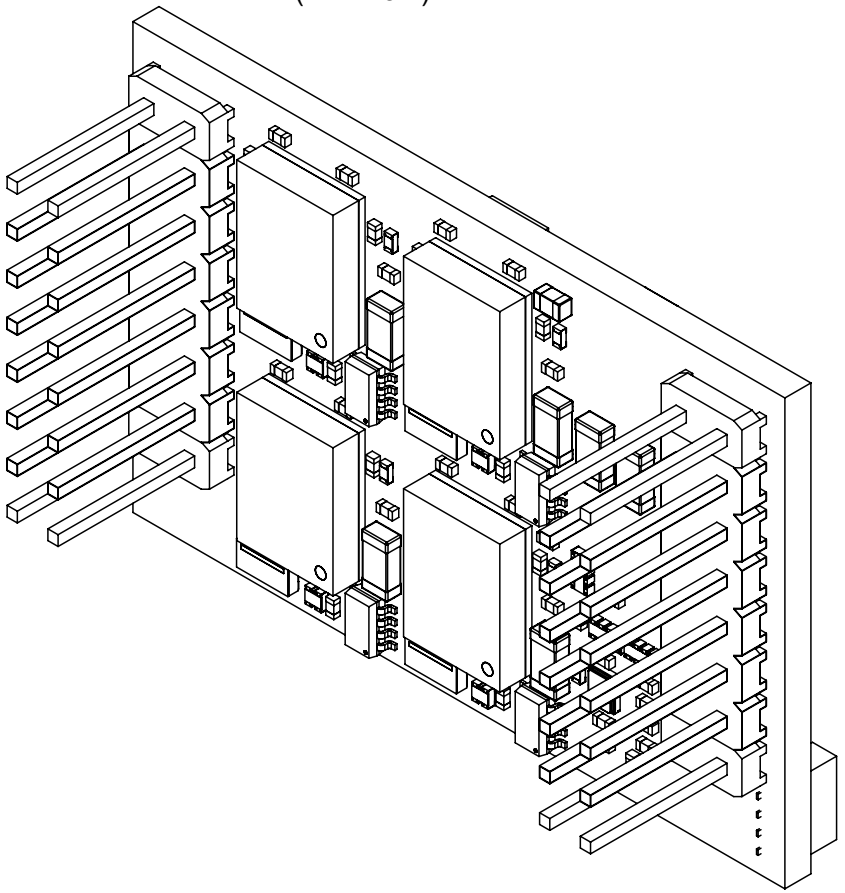
View from Bottom side (Scale 3.4:1)



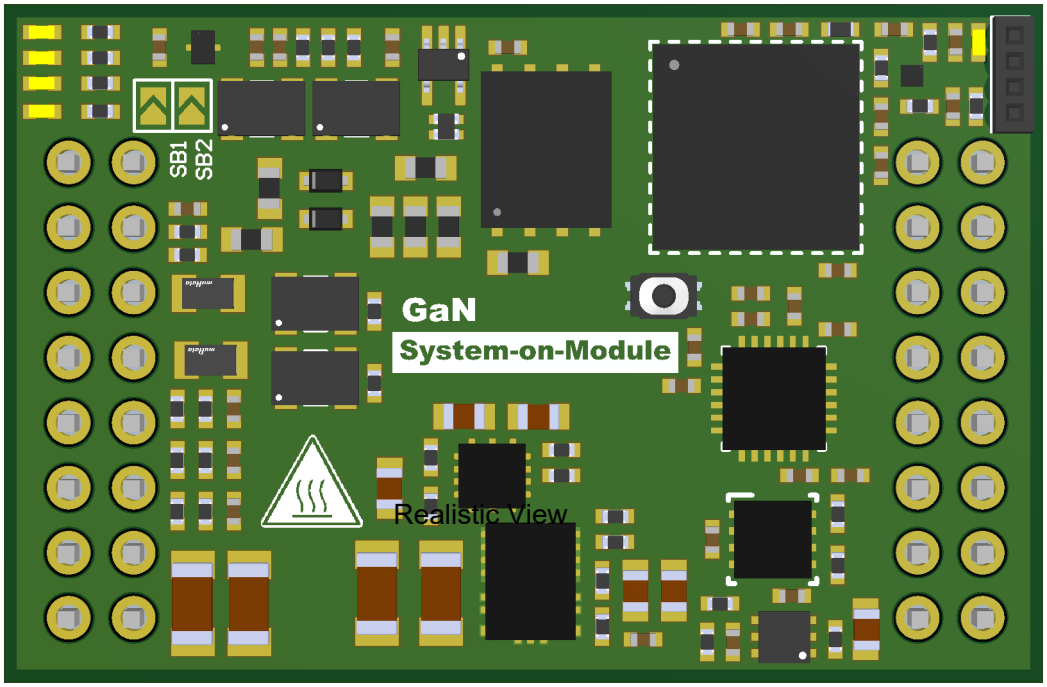
View from Top side (Scale 3:1)



View from Bottom side (Scale 3:1)



Realistic View



Realistic View

